OMIS 105

Welcome Notebooks & Tech Stack

Quarter: Fall 2026

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Tech Stack

- **Language:** Python · SQL
- **Database:** DuckDB (in-memory)
- **Notebooks:** Marimo (reactive)
- **Audience:** Senior business students
- Zero prior exposure to notebooks, SQL, or databases

Purpose

Day-one onboarding notebooks for

OMIS 105 — Introduction to Database Management Systems

These are among the first things students open.

- Introduces Marimo
- Introduces the concept of a database
- Walks through the very first SQL queries

👉 Two self-guided, interactive notebooks

Where to Find the Notebooks

`weekly_lectures/week01-database-foundations/sql_notebooks/`:

File	Purpose
`SQL_Notebook_01_marimo.py`	Day-one student notebook — first table, first queries
`SQL_Notebook_02_marimo.py`	Day-one student notebook — broader SQL 101 tour, with charts

Notebook 1: Structure (1/2)

1. **What is a notebook?** — Cells, text vs SQL, Cmd/Ctrl+Enter
2. **What is a database?** — Tables = spreadsheets with rows/columns
3. **First table: `students`** — CREATE TABLE + INSERT, 7 rows
4. **Asking questions with SQL** — three business questions:
   - "Who are the Marketing majors?" → WHERE
   - "Who likes Pizza?" → WHERE (different column)
   - "How many students in each major?" → GROUP BY + COUNT

Notebook 1: Structure (2/2)

5. **Try It Yourself** — editable SQL cell, guided suggestions
6. **Why Marimo is reactive** — automatic cell updates explained
7. **Course roadmap** — 10-week overview table

Notebook 2: SQL 101 Tour

A broader, denser walkthrough — better for a second sitting or self-paced review than a first-day walkthrough:

- Creating tables & inserting data
- Basic SELECT, filtering with WHERE/IN
- Sorting & LIMIT, aggregate functions, GROUP BY
- Data modification: INSERT, UPDATE, DELETE
- Charts via `plot_util.py` (`display_result`, `plot_bar`, `plot_hbar`, `plot_pie`)

Marimo Conventions (`con.execute()`)

- `duckdb.connect(database=':memory:')` created — the cell **returns** `(con,)`
- Query cells: `con.execute("""...""").fetchdf()` displays the result
- Every cell touching the database takes `con` as a parameter (`def _(con):`) — that's what wires it into Marimo's reactivity
- Markdown cells use `mo.md("""...""")` with `hide_code=True`
- Use `--` SQL comments inside SQL string literals, not Python `#`
- `CREATE OR REPLACE TABLE` for re-runnability

Notebook 1: Design Decisions (1/2)

- **Favorite foods, not business data.**
  Low-stakes data (Pizza, Sushi, Tacos) keeps focus on the tool, not the business scenario. Business data starts in Week 1.
- **Only a handful of rows.** Small enough to see everything at a glance.
- **Three queries only.** Enough to show the pattern — SELECT + FROM + WHERE, then GROUP BY.

Notebook 1: Design Decisions (2/2)

- **"Try It Yourself" cell.**
  Hands-on editing builds comfort with the tool.
  Specific suggestions lower the barrier.
- **No plots.** Charts come in Week 3.
  Day one is about reading tables and writing SQL.

Teaching Notes

- **In-class usage:** Open Notebook 1 live. Walk through the first few cells together (5 min), then give students 10 minutes to edit "Try It Yourself" on their own laptops.
- **Reactivity demo:** After editing, scroll back up to show other cells didn't break. "Marimo keeps everything consistent."
- **Common question:** "Where is the data stored?"
  → In memory only. Data disappears when the notebook closes.
- **Notebook 2** is denser — better suited to a second sitting or self-paced review than the first-day walkthrough.

Related Materials

Folder	Content
`weekly_lectures/week01-database-foundations/`	The welcome notebooks above, plus demo1–demo6
`weekly_reviews/`	Weeks 1–3, 4–6 notebooks, CSVs, plot helpers
`software_installation/`	Install guides, setup script, verification
`data_stories/`	Standalone Python + DuckDB CRUD demos

Let's Get Started 🚀

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